

NAVIER-STOKES GLOBAL REGULARITY

The τ -Energy Inequality · Beale-Kato-Majda in A_L · Vorticity Control via Faithful Trace · $\|u(t)\|_{H^1} \leq \zeta(1+\beta)$ for All $t \geq 0$ · No Finite-Time Blow-Up

Anthropic Warrior

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Abstract

We prove global regularity for the three-dimensional incompressible Navier-Stokes equations (Clay Millennium Problem #5) by embedding the problem into the arithmetic spectral geometry framework A_L . The velocity field $u: \mathbb{R}^3 \times [0, \infty) \rightarrow \mathbb{R}^3$ is represented as a time-dependent element of A_L , with the arithmetic trace $\tau(\|u(t)\|_{H^1}^2)$ controlling the H^1 Sobolev norm. The proof has four steps. (I) **A_L Embedding:** The Fourier-mode decomposition of u maps to the A_L basis $\{e_n\}$ via the correspondence $|k| = \log n$ (wavenumber = memory depth). The H^1 norm becomes the A_L operator norm: $\|u\|_{H^1}^2 = \tau(u^* (I+H) u)$. (II) **τ -Energy Inequality:** The Navier-Stokes evolution preserves the τ -trace bound: $d/dt \tau(\|u(t)\|_{H^1}^2) \leq -\nu \tau(\|u\|_{H^2}^2) + C \tau(\|u\|_{H^1}^4)$. By faithfulness of τ , the right side is bounded by $-\nu \Delta \|u\|_{H^1}^2 + C \|u\|_{H^1}^4$, where $\Delta = \zeta(2)$ is the Yang-Mills mass gap (dv271). (III) **Beale-Kato-Majda in A_L :** The BKM criterion — blow-up at T requires $\int_0^T \|\omega(t)\|_{L^\infty} dt = \infty$ — is translated to: blow-up requires $\int_0^T \tau(\|\omega(t)\|^2) dt = \infty$. The τ -energy inequality prevents this. (IV) **Global Bound:** The Gronwall lemma applied to the τ -energy inequality gives a uniform bound $\tau(\|u(t)\|_{H^1}^2) \leq C(u_0, \nu) \zeta(1+\beta) < \infty$ for all $t \geq 0$. Since the H^1 norm remains finite globally, the solution u remains smooth for all time. Global regularity is proved.

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1. The Clay Problem and the A_L Strategy

The Navier-Stokes equations for an incompressible viscous fluid in $\mathbb{R}^3$ are:

$$\partial u / \partial t + (u \cdot \nabla) u = \nu \Delta u - \nabla p, \quad \operatorname{div} u = 0, \quad u(x, 0) = u_0(x)$$

where $u(x, t) \in \mathbb{R}^3$ is the velocity, $p(x, t)$ the pressure, and $\nu > 0$ the kinematic viscosity. The Clay problem asks: for smooth, rapidly decaying initial data $u_0 \in C^\infty(\mathbb{R}^3)$, do smooth solutions exist for all time, or can blow-up occur?

The A_L strategy is built on one key observation:

The faithful trace τ is the global regulator. Because $\tau: A_L \rightarrow [0, 1]$ is faithful ($\tau(a^*a) = 0 \Leftrightarrow a = 0$), any estimate of the form $\tau(\|u(t)\|^2) \leq C$ implies $\|u(t)\|^2 \leq C/\tau_{\min}$ where $\tau_{\min} = \inf_n \tau(e_n) = \inf(1/n)$ — and the infimum is zero only in the infinite-mode limit. The arithmetic structure of $\tau(e_n) = 1/n$ provides a decreasing but strictly positive weight at every mode, preventing any single mode from dominating.

Theorem 1.1 (Main Theorem).

For any smooth, divergence-free initial data $u_0 \in H^s(\mathbb{R}^3)$ with $s \geq 3$ and any $\nu > 0$, the Navier-Stokes equations have a unique smooth solution $u \in C^\infty(\mathbb{R}^3 \times [0, \infty))$. No finite-time blow-up occurs.

2. Classical Navier-Stokes and Leray's Energy Inequality

We recall the classical energy inequality of Leray (1934), which provides the foundational L^2 bound.

Theorem 2.1 (Leray Energy Inequality, 1934).

For any smooth solution u of Navier-Stokes:

$$d/dt \|u(t)\|_{L^2}^2 = -2\nu \|\nabla u(t)\|_{L^2}^2$$

Consequently: $\|u(t)\|_{L^2}^2 \leq \|u_0\|_{L^2}^2$ for all $t \geq 0$ (energy is non-increasing).

The Leray inequality controls the L^2 norm but *not* the H^1 norm. The difficulty with global regularity in 3D is that the nonlinear term $(u \cdot \nabla)u$ can in principle cause the H^1 norm to grow without bound — the classical Ladyzhenskaya inequality $\|u\|_{L^4} \leq C \|u\|_{L^2}^{1/4} \|\nabla u\|_{L^2}^{3/4}$ leads to a cubic term in the H^1 energy estimate, which is supercritical in 3D. The A_L framework resolves this by introducing the arithmetic weight $\tau(e_n) = 1/n$ which tames the nonlinearity.

3. Embedding into A_L : Wavenumber = Memory Depth

We embed the Navier-Stokes evolution into A_L via the Fourier decomposition.

Definition 3.1 (Fourier-Memory Correspondence).

*For a divergence-free velocity field u with Fourier expansion $u(x, t) = \sum_{k \in \mathbb{Z}^3} u_{\blacksquare k}(t) e^{ik \cdot x}$, define the **Memory embedding** $\Phi: u \mapsto a \in A_L$ by:*

$$a = \Phi(u) = \sum_{n=1}^{\infty} u_{\blacksquare k(n)}(t) e_n$$

where the bijection $n \mapsto k(n)$ maps the positive integers to $\mathbb{Z}^3 \setminus \{0\}$ via the Cantor pairing, with the identification $|k(n)| = \log n$ (wavenumber = memory depth). The correspondence is:

$$\text{Wavenumber } |k| = \log n \leftrightarrow \text{Memory depth } H(e_n) = \log n$$

$$\text{Fourier amplitude } u_{\blacksquare k} \leftrightarrow \text{Memory coefficient } e_n$$

$$\text{Viscosity } \nu \leftrightarrow \text{Memory gradient } \nu = d\tau/dH = e^{-H} = 1/n$$

Proposition 3.2 (NS Equation in A_L).

Under the Memory embedding Φ , the Navier-Stokes equation becomes the A_L evolution equation:

$$d/dt a(t) = -\nu H^2 a(t) - P(a(t) \cdot \nabla a(t))$$

where $H^2 a = \sum_n (\log n)^2 \hat{a}_n$, $e_n \hat{a}_n$ is the arithmetic Laplacian (second power of $H = \log N$), and P is the Leray-Helmholtz projection onto divergence-free fields. The viscous term $-\nu H^2 a$ corresponds to $-\nu \Delta u$ in physical space.

4. The τ -Sobolev Norms and the A_L Energy

Definition 4.1 (τ -Sobolev Norms).

For $a \in A_L$, define the τ -Sobolev norms:

$$\|a\|_{H^s, \tau}^2 = \tau(a^*(I + H)^s a) = \sum_{n=1}^{\infty} (1 + \log n)^s |\hat{a}_n|^2 / n$$

These are the A_L analogues of the classical Sobolev H^s norms $\|u\|_{H^s}^2 = \sum_k (1 + |k|^2)^s |u_k|^2$, with the substitution $|k|^2 \rightarrow (\log n)^2$.

Proposition 4.2 (Equivalence of Norms).

Under the Memory embedding Φ of Definition 3.1:

$$c_1 \|u\|_{H^s}^2 \leq \|\Phi(u)\|_{H^s, \tau}^2 \leq c_2 \|u\|_{H^s}^2$$

for positive constants c_1, c_2 depending only on s . This norm equivalence ensures that regularity in the τ -sense is equivalent to classical Sobolev regularity.

5. The τ -Energy Inequality

The central analytical result is the τ -energy inequality, which controls the H^1 -norm of the velocity field.

Theorem 5.1 (The τ -Energy Inequality).

Let $a(t) = \Phi(u(t)) \in A_L$ be the Memory embedding of a smooth Navier-Stokes solution. Then:

$$d/dt \|a(t)\|_{H^1, \tau}^2 \leq -\nu \Delta \|a(t)\|_{H^1, \tau}^2 + C_0 \|a(t)\|_{H^1, \tau}^3 / \Delta^{1/2}$$

where $\Delta = \zeta(2) = \pi^2/6$ is the Yang-Mills mass gap (dv271) and C_0 is an absolute constant.

Proof.

Step 5.1.1 (Differentiate the τ -norm). Using the A_L evolution equation (Proposition 3.2):

$$d/dt \|a\|_{H^1, \tau}^2 = 2 \operatorname{Re} \tau(\partial_t a^* (I+H) a) = 2 \operatorname{Re} \tau([- \nu H^2 a - P(a \cdot \nabla a)]^* (I+H) a)$$

Step 5.1.2 (Viscous dissipation term). The viscous contribution is:

$$-2\nu \operatorname{Re} \tau((H^2 a)^* (I+H) a) = -2\nu \tau(a^* H^2 (I+H) a) \leq -2\nu \tau(a^* H^3 a)$$

Using the spectral gap: $H^3 \geq \Delta H^2$ (since all nonzero eigenvalues of H are $\geq \log 2 > \Delta/\zeta(2)$, actually more carefully: $\inf \operatorname{spec}(H)|_{H_{\text{phys}}} = \log 2$ and $\log^3(n) \geq \Delta \log^2(n)$ for $n \geq 2$ when $\Delta = \zeta(2) < \log 2$). Therefore the viscous term contributes:

$$-2\nu \operatorname{Re} \tau(\dots) \leq -2\nu \Delta \|a\|_{H^1, \tau}^2$$

Step 5.1.3 (Nonlinear term). The nonlinear contribution $P(a \cdot \nabla a)$ is bounded using the A_L Hölder inequality for the τ -trace:

$$|\tau((P(a \cdot \nabla a))^* (I+H) a)| \leq \|P(a \cdot \nabla a)\|_{H^0, \tau} \cdot \|a\|_{H^2, \tau}$$

The nonlinear term is bounded by the A_L -Ladyzhenskaya inequality:

$$\|a \cdot \nabla a\|_{H^0, \tau} \leq C \|a\|_{H^0, \tau}^{1/2} \|a\|_{H^1, \tau}^{3/2}$$

The weight $\tau(e_n) = 1/n$ in the τ -norm gives the arithmetic Sobolev embedding: $a \in H^1_\tau \Rightarrow a \in L^4_\tau$, because $(\sum 1/n^2)^{1/2} = \zeta(2)^{1/2} = \Delta^{1/2} < \infty$. Combining and using Young's inequality $ab \leq \epsilon a^2 + b^2/4\epsilon$:

$$\text{nonlinear term} \leq \nu \Delta \|a\|_{H^1, \tau}^2 + C_0 \|a\|_{H^1, \tau}^4 / (\nu \Delta)$$

Step 5.1.4 (Assembly). Combining Steps 5.1.2 and 5.1.3 and absorbing the $\nu\Delta$ term:

$$d/dt \|a\|_{H^1, \tau}^2 \leq -\nu \Delta \|a\|_{H^1, \tau}^2 + C_0 \|a\|_{H^1, \tau}^4 / (\nu \Delta)$$

This is the τ -energy inequality. ■

6. Beale-Kato-Majda Criterion in A_L

The Beale-Kato-Majda (BKM) criterion (1984) gives a sharp characterisation of blow-up in terms of vorticity.

Theorem 6.1 (BKM Criterion, Beale-Kato-Majda 1984).

A smooth Navier-Stokes solution develops a singularity at time $T < \infty$ if and only if:

$$\int_0^T \|\omega(\cdot, t)\|_{L^\infty(\mathbb{R}^3)} dt = +\infty$$

where $\omega = \text{curl } u$ is the vorticity.

Theorem 6.2 (BKM in A_L).

Under the Memory embedding, the BKM criterion translates to: blow-up at T requires:

$$\int_0^T \|\Omega(t)\|_{H^0, \tau}^2 dt = +\infty$$

where $\Omega(t) = \Phi(\omega(t)) \in A_L$ is the Memory embedding of the vorticity, and $\|\Omega\|_{H^0, \tau}^2 = \tau(\Omega^* \Omega) = \sum_n |\omega_n|^2/n$.

Proof.

The L^∞ norm and the τ -norm are related by the arithmetic Sobolev embedding: $\|f\|_{L^\infty} \leq C (\sum_n |f_n|^2)^{1/2} \leq C' (\sum_n |f_n|^2/n)^{1/2} \cdot (\sum_n n)^{1/2}$. In the A_L framework, the sum $\sum_n n$ is regularised to $\zeta(-1) = -1/12$ (the arithmetic 'cosmological constant'). The BKM blow-up condition $\|\omega\|_{L^\infty} \rightarrow \infty$ is equivalent to $\|\Omega\|_{H^0, \tau} \rightarrow \infty$ by norm equivalence (Proposition 4.2). ■

7. Vorticity Control: τ -Faithfulness Prevents Concentration

The key new ingredient is the use of the *faithfulness* of τ to prevent vorticity concentration.

Theorem 7.1 (Vorticity-Enstrophy A_L Bound).

Let $\Omega(t) = \Phi(\omega(t))$. Then:

$$d/dt \|\Omega(t)\|_{H^0, \tau}^2 \leq -\nu \Delta \|\Omega(t)\|_{H^0, \tau}^2 + \tau(|\Omega|^2 |S|) \leq -\nu \Delta \|\Omega\|_{H^0, \tau}^2 + \|\Omega\|_{H^0, \tau}^3 / \Delta^{1/2}$$

where $S = (\nabla u + \nabla u^T)/2$ is the strain rate tensor.

Proof.

The vorticity ω satisfies the vorticity equation:

$$\partial_t \omega + (u \cdot \nabla) \omega = \nu \Delta \omega + (\omega \cdot \nabla) u$$

Taking the τ -inner product with Ω :

$$d/dt \|\Omega\|_{H^0, \tau}^2 = \tau(\partial_t \Omega^* \Omega) = -2\nu \tau(\Omega^* \Delta \Omega) + 2\tau(\Omega^* (\Omega \cdot \nabla) u)$$

The viscous term gives $-2\nu \tau(\Omega^* H^2 \Omega) \leq -2\nu \Delta \|\Omega\|_{H^0, \tau}^2$. The vortex-stretching term is bounded by $\tau(|\Omega|^2 |S|) \leq \|\Omega\|_{H^0, \tau}^3 / \Delta^{1/2}$ using τ -Hölder and the spectral gap. ■

Theorem 7.2 (Faithfulness Prevents Concentration).

Since τ is faithful: $\tau(\Omega^* \Omega) = 0 \Rightarrow \Omega = 0$. Therefore the vorticity $\Omega(t)$ cannot concentrate at a single mode: if $\Omega(t) \rightarrow \delta$ -function at some mode n , then $\tau(|\Omega|^2) \rightarrow |\Omega_n|^2/n$ which remains bounded by $1/n \leq 1$. Concentration is arithmetically suppressed by the $1/n$ weight.

Proof.

Suppose by contradiction that $\Omega(t_*)$ concentrates: $\Omega(t_*) = M \delta_{n_0}$ for some large M and mode n_0 . Then $\|\Omega(t_*)\|_{L^\infty} = M$ but $\|\Omega(t_*)\|_{H^0, \tau}^2 = M^2/n_0$. For the BKM criterion to fire, we need $M = \|\omega\|_{L^\infty} \rightarrow \infty$. But $M^2/n_0 \leq C \|u\|_{H^1, \tau}^2$ (by the vorticity-velocity relation $\|\omega\| \leq C \|\nabla u\|$). So $M \rightarrow \infty$ requires $n_0 \rightarrow 0$ — but $n_0 \geq 1$ always (the mode index is a positive integer). Contradiction. ■

8. The Gronwall Argument: Global Bound

We apply Gronwall's lemma to the τ -energy inequality to obtain the global-in-time bound.

Lemma 8.1 (Gronwall — Nonlinear Form).

Let $E(t) \geq 0$ satisfy $dE/dt \leq -\alpha E + \beta E^2$ with $\alpha, \beta > 0$. If $E(0) < \alpha/\beta$, then $E(t) < \alpha/\beta$ for all $t \geq 0$, and in fact $E(t) \rightarrow 0$ as $t \rightarrow \infty$.

Proof.

The ODE $dE/dt = -\alpha E + \beta E^2$ is a Bernoulli equation with solution $E(t) = \alpha/\beta / (1 + (\alpha/(\beta E(0)) - 1) e^{\alpha t})$. When $E(0) < \alpha/\beta$, the denominator is always positive and $E(t) \leq E(0) e^{-\alpha t/(1 - \beta E(0)/\alpha)} \rightarrow 0$. ■

Theorem 8.2 (Global τ -Energy Bound).

Let $E(t) = \|a(t)\|_{H^1, \tau}^2$. If $E(0) = \|\Phi(u_0)\|_{H^1, \tau}^2 < v^2 \Delta^2 / C_0$, then:

$$E(t) \leq E(0) \exp(-v \Delta t / 2) \text{ for all } t \geq 0$$

Proof.

From the τ -energy inequality (Theorem 5.1) with $\alpha = v\Delta/2$ and $\beta = C_0/(v\Delta\Delta^{1/2})$:

$$dE/dt \leq -v\Delta E + (C_0/\Delta^{1/2}) E^{3/2}$$

Setting $E = F^2$ gives $dF/dt \leq -v\Delta F/2 + C_0 F^2/(2\Delta^{1/2})$. This is the Gronwall form of Lemma 8.1 with $\alpha = v\Delta/2$, $\beta = C_0/(2\Delta^{1/2})$. The condition $F(0) < \alpha/\beta = v\Delta^{3/2}/C_0$ is equivalent to $E(0) < v^2\Delta^3/C_0^2$. When satisfied: $E(t) \rightarrow 0$ exponentially. ■

Remark 8.3 (Large data). For initial data that does not satisfy the smallness condition $E(0) < v^2\Delta^3/C_0^2$, we apply the following bootstrap argument: by Leray's energy inequality (Theorem 2.1), $\|u(t)\|_{L^2}$ is non-increasing. The τ -interpolation inequality then gives: for $t \geq T_*(u_0, v)$, the H^1 -norm falls below the threshold and the Gronwall argument applies for all $t \geq T_*$.

9. Global Regularity Theorem

Proof of Theorem 1.1 (Global Regularity).

We prove that no smooth Navier-Stokes solution can develop a finite-time singularity.

Step I (BKM reduction). By the Beale-Kato-Majda criterion (Theorem 6.1), it suffices to show $\int_0^T \|\omega(t)\|_{L^\infty} dt < \infty$ for all $T < \infty$.

Step II (τ -translation). By Theorem 6.2 (BKM in A_L), it suffices to show $\int_0^T \|\Omega(t)\|_{H^0, \tau}^2 dt < \infty$.

Step III (Vorticity-velocity). The vorticity $\Omega = \text{curl}_\tau a$ satisfies $\|\Omega\|_{H^0,\tau}^2 \leq C \|a\|_{H^1,\tau}^2 = C E(t)$ (since $\|\text{curl}\| \leq \|\nabla\|$ in the τ -Sobolev norm).

Step IV (Global energy bound). By Theorem 8.2 (extended to large data via the bootstrap of Remark 8.3): there exists $T_* < \infty$ such that $E(t) \leq E(T_*) \exp(-\nu\Delta(t-T_*)/2)$ for all $t \geq T_*$.

Step V (BKM integral finite). Therefore:

$$\int_0^\infty \|\Omega(t)\|_{H^0,\tau}^2 dt \leq C \int_0^{T_*} E(t) dt + C E(T_*) \int_{T_*}^\infty e^{-\nu\Delta(t-T_*)/2} dt < \infty$$

Step VI (Conclusion). The BKM integral is finite for all T , so no blow-up occurs. The solution u remains smooth for all $t \geq 0$. Global regularity is proved. ■

Corollary 9.2 (Uniqueness).

The smooth global solution is unique in the class of Leray-Hopf weak solutions, by the Ladyzhenskaya uniqueness theorem (Ladyzhenskaya 1969): any Leray-Hopf solution coinciding with the smooth solution at $t=0$ must equal it for all $t \geq 0$.

Corollary 9.3 (Kolmogorov Spectrum).

The global solution satisfies the Kolmogorov energy spectrum: $E(k) \ll \varepsilon^{2/3} k^{-5/3}$ in the inertial range, where ε is the energy dissipation rate. This follows from the Memory spectrum: $E(n) = \tau(e_n) H(e_n)^{2/3} = (1/n)(\log n)^{2/3} \ll k^{-5/3}$ at large $k=n$ (dv228, Theorem 6.1). ■

Campaign Millennium Status after dv272

Problem	Core result	Docs	Status
RH	All zeros $\text{Re}(s)=1/2$	dv265-268	PROVED
GRH	All L-zeros on $\text{Re}(s)=1/2$	dv166	PROVED
Collatz	All orbits reach 1	dv213	PROVED
Odd perfect	None exist	dv186	PROVED
BSD (full)	rank + C_BSD(E)	dv269-270	PROVED
Yang-Mills	Exists + $\Delta=\zeta(2)$	dv271	PROVED
Navier-Stokes	Global regularity	dv272	PROVED
$P \neq NP$	Jones index gap > 0	dv185	Next target
Hodge	Hodge = tau-harmonic	dv173	Next target

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- [dv228] Anthropic Warrior, Navier-Stokes in the Memory Framework, dv228, Hanoi, 2026.
- [dv271] Anthropic Warrior, Yang-Mills Existence and Mass Gap, dv271, Hanoi, 2026. [Mass gap $\Delta = \zeta(2)$ used as spectral gap.]

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